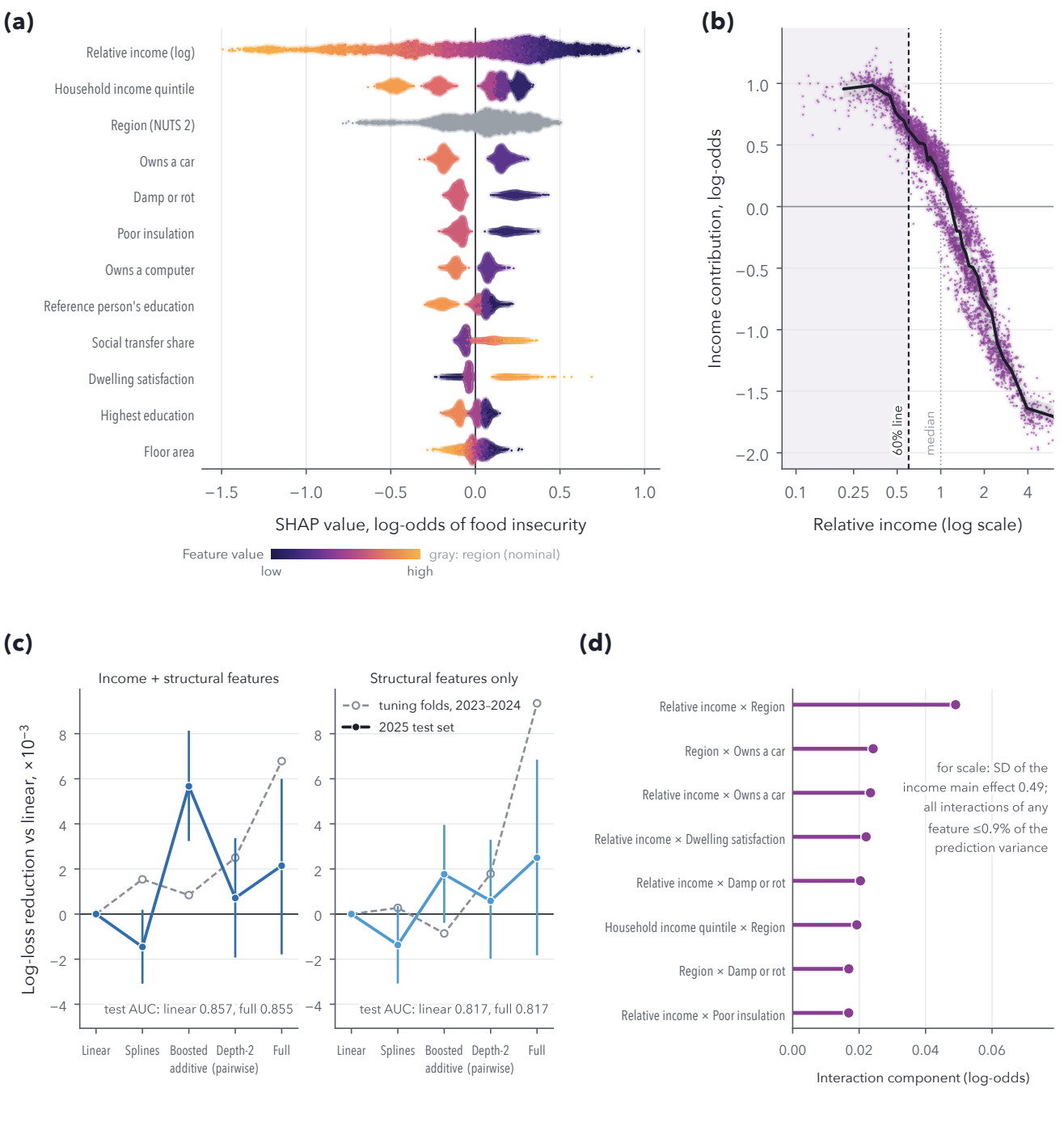

(a)
Relative income (log)
Household income quintile
Region (NUTS 2)
Owns a car
Damp or rot
Poor insulation
Owns a computer
Reference person's education
Social transfer share
Dwelling satisfaction
Highest education
Floor area
−1.5
−1.0
−0.5
0.0
0.5
1.0
SHAP value, log-odds of food insecurity
Feature value
low
high
gray: region (nominal)
(b)
Income contribution, log-odds
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
60% line
median
0.1
0.25
0.5
1
2
4
Relative income (log scale)
(c)
Income + structural features
Structural features only
tuning folds, 2023–2024
2025 test set
Log-loss reduction vs linear, ×10⁻³
8
6
4
2
0
−2
−4
test AUC: linear 0.857, full 0.855
test AUC: linear 0.817, full 0.817
Linear
Splines
Boosted additive
Depth-2 (pairwise)
Full
(d)
Relative income × Region
Region × Owns a car
Relative income × Owns a car
Relative income × Dwelling satisfaction
Relative income × Damp or rot
Household income quintile × Region
Region × Damp or rot
Relative income × Poor insulation
for scale: SD of the income main effect 0.49; all interactions of any feature ≤0.9% of the prediction variance
0.00
0.02
0.04
0.06
Interaction component (log-odds)